

Direct Transfer: Classic Approach

1. Morphological analysis as pre-processing:
 - “Mary didn’t slap the green witch” → “Mary DO:PAST not slap the green witch”
 - "מרי לא סטרה למכשפה הירוקה" →
"מרי לא סטר:עבר, נקבה ל מכשפה ה ירוק:נקבה"
2. Lexical Transfer
 - Mary DO:PAST not slap the green witch →
Maria no dar:PAST una bofetada a la verde bruja
3. Lexical Reordering
 - Maria no dar:PAST una bofetada a la bruja verde
4. Morphological generation
 - Maria no dió una bofetada a la bruja verde

Syntactic Transfer: Classic Approach

- Simple lexical reordering does not adequately handle more dramatic reordering such as that required to translate from an SVO to an SOV language.
- Need syntactic transfer rules that map parse tree for one language into one for another:
 - English to Spanish:
 - $NP \rightarrow Adj\ Nom \Rightarrow NP \rightarrow Nom\ ADJ$
 - English to Japanese:
 - $VP \rightarrow V\ NP \Rightarrow VP \rightarrow NP\ V$
 - $PP \rightarrow P\ NP \Rightarrow PP \rightarrow NP\ P$